


# Certified Penetration Testing Professional

## Methodology: IoT Penetration Testing

| | | | |
|---|---|---|---|
| **Penetration Tester:** | | | |
| **Organization:** | | | |
| **Date:** | | **Location:** | |


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## Test 1: IoT Penetration Testing

### Test 1.1: Attack Surface Mapping

| | |
|---|---|
| **Target Organization** | |
| **URL** | |
| **Target IoT Device** | |
| **Tests Performed** | 1.<br>2.<br>3.<br>4.<br>5. |

| | | |
|---|---|---|
| **Attack Surface Mapping Successful?** | ☐ Yes | ☐ No |

**Identified Attack Surfaces**

| | Attack Surface | Criticality | Nature of Exploitation |
|---|---|---|---|
| 1. | | | |
| 2. | | | |
| 3. | | | |
| 4. | | | |
| 5. | | | |

| | |
|---|---|
| **Tools/Services Used** | 1.<br>2.<br>3.<br>4.<br>5. |


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| Results Analysis: |
| --- |
| |


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## Test 1.2: Analyzing the IoT Hardware

| | |
|---|---|
| **Target Organization** | |
| **URL** | |
| **Target IoT Device** | |
| **Step-1: Research the Device** | |
| **Information obtained through Research** | 1.<br>2.<br>3.<br>4.<br>5. |
| **Step-2: Identify the Components** | |
| **Components of the IoT Device** | 1.<br>2.<br>3.<br>4.<br>5. |
| **Step-3: Identify the Debugging Ports** | |
| **Debugging Ports of the IoT Device** | 1.<br>2.<br>3.<br>4.<br>5. |
| **Step-4: Dump the Flash** | |
| **Information Extracted from Memory Dump** | 1.<br>2.<br>3. |


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| | |
|---|---|
| | 4.<br>5. |
| **Step-5: Extract/Analyze the firmware** | |
| **Results of Firmware Analysis** | 1.<br>2.<br>3.<br>4.<br>5. |
| **Identified Vulnerabilities** | 1.<br>2.<br>3.<br>4.<br>5. |
| **Tools/Services Used** | 1.<br>2.<br>3.<br>4.<br>5. |

| **Results Analysis:** |
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## Test 1.3: Attack Surface Map

| | | | | |
|---|---|---|---|---|
| **Target Organization** | | | | |
| **URL** | | | | |
| **Target IoT Device** | | | | |
| **Attack Surface Mapping Successful?** | | | ☐ Yes | ☐ No |
| **Information Collected** | | | | |
| | **FCC ID** | **Components** | **Attack Vectors** | **Attack Vector Rating** |
| 1. | | 1.<br>2.<br>3. | 1.<br>2.<br>3. | |
| 2. | | 1.<br>2.<br>3. | 1.<br>2.<br>3. | |
| 3. | | 1.<br>2.<br>3. | 1.<br>2.<br>3. | |
| 4. | | 1.<br>2.<br>3. | 1.<br>2.<br>3. | |
| 5. | | 1.<br>2.<br>3. | 1.<br>2.<br>3. | |
| 6. | | 1.<br>2.<br>3. | 1.<br>2.<br>3. | |


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| Tools/Services Used | 1.<br>2.<br>3.<br>4.<br>5. |
|---|---|

| Results Analysis: |
|---|
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## Test 1.4: Binwalk to Extract the File System

| | |
|---|---|
| **Target Organization** | |
| **URL** | |
| **Target IoT Device** | |
| **Commands Used** | |
| **Extracted File System Successfully?** | ☐ Yes ☐ No |

| **Data Obtained** | | | |
|---|---|---|---|
| | **Decimal** | **Hexadecimal** | **Description** |
| 1. | | | |
| 2. | | | |
| 3. | | | |
| 4. | | | |
| 5. | | | |
| 6. | | | |
| 7. | | | |
| 8. | | | |
| 9. | | | |
| 10. | | | |

| | |
|---|---|
| **Tools/Services Used** | 1. _____<br>2. _____<br>3. _____<br>4. _____<br>5. _____ |


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| Results Analysis: |
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## Test 1.5: Explore the File System

| | | |
|---|---|---|
| **Target Organization** | | |
| **URL** | | |
| **Target IoT Device** | | |
| **Commands Used** | | |
| **Explored the File System Successfully?** | ☐ Yes | ☐ No |
| **File System Type** | ☐ Squashfs<br>☐ Cramfs<br>☐ JFFS2<br>☐ YAFFS2<br>☐ ext2<br>☐ Any Other, Specify_____ | |
| **Compression Type** | ☐ LZMA<br>☐ Gzip<br>☐ Zip<br>☐ Zlib<br>☐ Arj<br>☐ Any Other, Specify_____ | |
| **Vulnerabilities Identified** | 1. _____<br>2. _____<br>3. _____<br>4. _____<br>5. _____ | |


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| Tools/Services Used | 1.<br>2.<br>3.<br>4.<br>5. |
|---|---|

| Results Analysis: |
|---|
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